

Prashanth B H

AI Engineer: Generative AI, Agentic AI & MLOps

Expertise: LLM Orchestration (LangChain) | Multi-Agent Systems (CrewAI, AutoGen)

RAG & MCP | Cloud MLOps (AWS, Azure, GCP)

+91-9880135087 | bhprashanth0@gmail.com | [LinkedIn](#) | [GitHub](#)

Summary:

AI Engineer with 6 years of combined experience in designing and deploying production-scale Generative AI, MLOps, and Machine Learning systems. Expertise spans multimodal AI (video, audio, text), LLM fine-tuning, Agentic RAG, and end-to-end MLOps pipelines. Proven track record in building scalable, low-latency APIs and containerized inference systems on cloud platforms (GCP, AWS, Azure) that drive measurable business impact. Strong foundation in software engineering, cross-functional collaboration, and mentoring.

Skill:

- Generative AI & LLMs: Prompt Engineering, LLM Fine-tuning (LoRA/QLoRA), RAG Pipelines, Agentic AI (LangGraph, CrewAI), Function Calling, Multimodal Transformers (CLIP), Model Evaluation (RAGAS).
- MLOps & Infrastructure: Kubeflow, TFX, MLflow, Weights & Biases, DVC, Airflow/Prefect, Docker, Kubernetes, Seldon Core/KServe, Triton Inference Server, Prometheus/Grafana.
- Cloud & AI Services: Vertex AI, GCP AI Platform, AWS SageMaker, AWS Bedrock, Azure ML, Azure OpenAI Service, Databricks.
- Frameworks & Libraries: PyTorch, TensorFlow, Transformers, LangChain, LlamaIndex, FastAPI, OpenCV, Scikit-learn, Pandas, NumPy.
- Data & Search: Vector Databases (Pinecone, Weaviate, FAISS, ChromaDB), Elasticsearch, Redis, MySQL, Hybrid Search.
- Core Programming: Python, Java, SQL, DSA.
- Concepts: Distributed Training (DeepSpeed), Model Quantization, GPU/CUDA Optimization, Microservices, CI/CD for ML, A/B Testing, Agile/Scrum.

Work Experience details:

Independent AI Engineer & Researcher

Dec 2022 - Present

- Designed and deployed end-to-end Generative AI systems: RAG-based Regulatory QA Bot, Agentic AI Assistant, multimodal AI pipelines.
- Built and containerized MLOps pipelines using Docker, FastAPI, MLflow, KServe, Prefect, and Ray.
- Implemented scalable AI applications on GCP, AWS, and Azure, supporting FastAPI backends and Streamlit/Gradio frontends.
- Developed LLM Evaluation & Benchmarking Platform using LangSmith, TruLens, and Arize AI to monitor accuracy, hallucinations, latency, and performance.
- Handled 10K+ documents and 100+ concurrent users for RAG systems; optimized inference costs by 35%.
- Mentored junior ML engineers, conducted technical interviews, and led architecture reviews.

Project 4: Fraud Detection System with Ensemble AI & Agentic Analysis

Jan 2026

Engineered an ensemble ML pipeline combining Isolation Forest and LSTM Autoencoder with custom scoring, achieving 92% detection rate and <0.8% false positives. Built LLM-augmented workflow using Azure OpenAI embeddings, RAG with FAISS, and Model Context Protocol (MCP) server for tool exposure. Implemented Agentic AI Copilot with CrewAI, featuring specialized agents with vector memory and function calling for automated investigation reports.

Impact: Reduced false positives by 60% and accelerated investigation cycles by 65% through automated AI analysis.

Technology Stack: Python, Isolation Forest, LSTM Autoencoder, Ensemble Learning, Azure OpenAI, RAG, Embeddings, Vector DB, CrewAI, Agentic AI, MCP (Server/Client), Copilot, FastAPI.

Project 3: Agentic AI System with Multi – Agent Orchestration & Tool Integration

Sep 2025 – Oct 2025

Designed and implemented a sophisticated agentic AI system using LangGraph, CrewAI, and AutoGen to orchestrate multi-agent workflows. The system integrates function calling for external APIs and databases, maintains context through vector and SQL memory, and applies the ReAct pattern for reasoning and task execution.

Impact: Built a state-of-the-art agent architecture capable of handling complex, multi-step tasks with persistent memory. Enabled seamless integration with external systems (APIs, databases) for real-world automation. Demonstrated advanced workflow orchestration and reasoning, illustrating production-ready AI agent design.

Technology: CrewAI, Function Calling, Vector Databases, SQL Agent, ReAct Pattern, MCP.

Project 2: End-to-End MLOps Platform for LLM Fine-Tuning

Apr 2025 – June 2025

Designed a platform to automate the **fine-tuning lifecycle of LLMs (Llama 2)** using **DVC for data versioning**, **MLflow for experiment tracking**, and **Kubeflow Pipelines** for orchestration.

Implemented **model serving with Seldon Core**, enabling **A/B testing** and **canary deployments**, with monitoring setup using **Prometheus** and **Grafana**.

Demonstrates: Full LLM MLOps lifecycle, cost-effective fine-tuning (LoRA), and production monitoring.

Project 1: Multi-Modal RAG System for Video Content

Nov 2024 – Dec 2024

Architected a **RAG pipeline for video transcripts, frames, and metadata** using **CLIP embeddings** for multimodal retrieval and **Weaviate** as a vector database.

Implemented **hybrid search** (vector + keyword) and **real-time video processing** features like scene detection and summarization for a media workflow application.

Demonstrates: Multimodal AI expertise, hybrid search implementation, and media industry-relevant AI solutions.

Final Year Project: Car Black Box System

- Engineered a vehicle data recorder system based on the PIC18F4580 microcontroller, designed to capture and log critical metrics. Integrated Implemented RTC using I2C protocol for precise timestamping. Implemented UART protocol for downloading stored data logs and connected a CLCD for real-time display. Utilized the microcontroller's 10-bit ADC to capture analog sensor data for metrics like vehicle speed.
- Impact:** Created a functional prototype that successfully captures, timestamps, and stores essential driving parameters, demonstrating core embedded systems design principles for data logging and diagnostics.
- Tech Stack:** PIC18F4580, Embedded C, I2C & UART Protocols, RTC Module, 10-bit ADC, CLCD, MPLAB X IDE.

Certifications

- Java Full Stack Developer | Pentagon Space | *2024*
- Programming in Java | NPTL | *2023*
- Data Science and Machine Learning | Inventeron Technologies | *2023*
- Deep Learning Specialization | Coursera (deeplearning.ai) | *2024*
- Generative AI with LLMs | Coursera (AWS & deeplearning.ai) | *2024*
- MLOps Fundamentals | Coursera | *2024*

Education:

Bachelor of Technology in Electronics & Communication
G M Institute of Technology, VTU, Karnataka | CGPA: 7.42/10

Languages - English (Intermediate) | Kannada (Native) | Hindi (Professional)